Chapter 3: Determinants

3.1 Introduction to Determinants

- We can discover the definition for the 3×3 case by watching what happens when an invertible 3×3 matrix A is row reduced.
- Consider $A=[a_{ij}]$ with $a_{11}\neq 0$. If we multiply the second and third rows of A by a_{11} and then subtract appropriate multiples of the first row from the other two rows, we find that A is row equivalent to the following two matrices.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{11}a_{21} & a_{11}a_{22} & a_{11}a_{23} \\ a_{11}a_{31} & a_{11}a_{32} & a_{11}a_{33} \end{bmatrix} \sim \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{11}a_{22}-a_{12}a_{21} & a_{11}a_{23}-a_{13}a_{21} \\ 0 & a_{11}a_{32}-a_{12}a_{31} & a_{11}a_{33}-a_{13}a_{31} \end{bmatrix} \quad (1)$$

- Since A is invertible, either the (2,2) entry or the (3,2) entry on the right in (1) is nonzero (since each row of A must have a pivot)
- Let us suppose that the (2,2) entry is nonzero. (Otherwise, we can make a row interchange before proceeding.)
- Multiply row 3 by $a_{11}a_{22}-a_{12}a_{21}$, and then to the new row 3 add $-(a_{11}a_{32}-a_{12}a_{31})$ times row 2. This will show that

$$A \sim \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{11}a_{22}-a_{12}a_{21} & a_{11}a_{23}-a_{13}a_{21} \\ 0 & 0 & a_{11}\Delta \end{bmatrix}$$

where $\Delta = a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32}-a_{11}a_{23}a_{32}-a_{12}a_{21}a_{33}-a_{13}a_{22}a_{31}$ (2)

Since A is invertible, Δ must be nonzero. The converse is true, too which we shall prove in Section 3.2. $\Delta \rightarrow$ **Determinant** of the 3×3 matrix A.